A Compute-First Lecture Note on Riemann–Roch for $\mathbb{CP}^1$ with 30 Exercises

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1 Purpose of this note

The goal of this note is to understand the Riemann–Roch theorem in the easiest nontrivial compact Riemann surface, namely

$$X = \mathbb{CP}^1.$$

On a general compact Riemann surface, Riemann–Roch says

$$\ell(D) - \ell(K - D) = \deg D + 1 - g,$$

where

- D is a divisor,
- K is a canonical divisor,
- g is the genus,
- $\ell(D) = \dim_{\mathbb{C}} L(D)$, where

$$L(D) := \{f \in \mathcal{M}(X) : (f) + D \geq 0\} \cup \{0\}.$$

For $X = \mathbb{CP}^1$ we have $g = 0$, so the theorem becomes

$$\ell(D) - \ell(K - D) = \deg D + 1.$$

The miracle on $\mathbb{CP}^1$ is that every divisor is linearly equivalent to a multiple of $[\infty]$, so the entire theorem reduces to explicit calculations with rational functions and polynomials.

The strategy of this note is:

1. understand coordinates on $\mathbb{CP}^1$,
2. compute divisors of rational functions,
3. show every divisor is equivalent to $n[\infty]$,
4. compute $L(n[\infty])$ explicitly,
5. compute the canonical divisor,
6. verify Riemann–Roch directly.

2 The surface $\mathbb{CP}^1$ and its coordinates

Recall

$$\mathbb{CP}^1 = (\mathbb{C}^2 \setminus \{(0,0)\})/\mathbb{C}^\times,$$

where

$$\lambda \cdot (Z_0, Z_1) = (\lambda Z_0, \lambda Z_1), \qquad \lambda \in \mathbb{C}^\times.$$

A point is written $[Z_0 : Z_1]$.

There are two standard affine charts:

$$U_0 = \{Z_0 \neq 0\}, \qquad U_\infty = \{Z_1 \neq 0\}.$$

On U_0 we use the coordinate

$$z = \frac{Z_1}{Z_0},$$

so $U_0 \cong \mathbb{C}$. On U_∞ we use the coordinate

$$w = \frac{Z_0}{Z_1},$$

so $U_\infty \cong \mathbb{C}$. On the overlap $U_0 \cap U_\infty$ we have

$$w = \frac{1}{z}.$$

Thus $\mathbb{CP}^1$ is obtained by gluing two copies of $\mathbb{C}$ by inversion.

The point at infinity is

$$\infty = [0 : 1],$$

which lies in U_∞ and corresponds to $w = 0$.

3 Meromorphic functions on $\mathbb{CP}^1$

A meromorphic function on $\mathbb{CP}^1$ is exactly a rational function in the affine coordinate z.

Proposition 3.1. *We have*

$$\mathcal{M}(\mathbb{CP}^1) = \mathbb{C}(z).$$

Proof. A rational function in z is certainly meromorphic on the z-chart and also meromorphic near ∞ after the substitution $z = 1/w$. Conversely, let f be meromorphic on $\mathbb{CP}^1$. Restrict f to $U_0 \cong \mathbb{C}$; then f is meromorphic on $\mathbb{C}$. Since ∞ is an isolated singularity of f, the function has at worst a pole at infinity. A meromorphic function on $\mathbb{C}$ with at worst a pole at infinity is rational. Hence $f \in \mathbb{C}(z)$. □

So every global meromorphic function is concrete: it is just a ratio $P(z)/Q(z)$ of polynomials.

4 Divisors and orders on $\mathbb{CP}^1$

Definition 4.1. A divisor on X is a finite formal sum

$$D = \sum_{p \in X} n_p[p], \qquad n_p \in \mathbb{Z}.$$

Its degree is

$$\deg D = \sum_{p \in X} n_p.$$

If f is a nonzero meromorphic function and $p \in X$, then $\operatorname{ord}_p(f)$ denotes the vanishing order of f at p (negative if f has a pole). The principal divisor of f is

$$(f) = \sum_{p \in X} \operatorname{ord}_p(f)[p].$$

Example 4.2. In the coordinate z on U_0, the function $z - a$ has a simple zero at $a \in \mathbb{C}$. To understand what happens at ∞, write $z = 1/w$. Then

$$z - a = \frac{1 - aw}{w}.$$

Near $w = 0$, the factor $1 - aw$ is a unit, while $1/w$ has a simple pole. Therefore

$$(z - a) = [a] - [\infty].$$

Example 4.3. More generally,

$$\left(\frac{z - a}{z - b}\right) = [a] - [b].$$

Proposition 4.4. *Every principal divisor on $\mathbb{CP}^1$ has degree 0.*

Proof. Let $f = P/Q$ with coprime polynomials. If $\deg P = m$ and $\deg Q = n$, then the finite zeros contribute m, the finite poles contribute $-n$, and near infinity we have

$$f(1/w) = \frac{P(1/w)}{Q(1/w)} \sim c\, w^{n-m},$$

so

$$\operatorname{ord}_\infty(f) = n - m.$$

Hence

$$\deg(f) = m - n + (n - m) = 0.$$

□

5 Linear equivalence on $\mathbb{CP}^1$

Two divisors D, E are linearly equivalent, written $D \sim E$, if $D - E = (f)$ for some nonzero meromorphic function f.

The key fact on $\mathbb{CP}^1$ is that degree is the only invariant of divisor class.

Theorem 5.1. *Every divisor D on $\mathbb{CP}^1$ is linearly equivalent to $(\deg D)[\infty]$. In particular,*

$$\operatorname{Pic}(\mathbb{CP}^1) \cong \mathbb{Z}, \qquad D \longmapsto \deg D.$$

Proof. Write

$$D = \sum_{i=1}^{r} m_i[a_i] - \sum_{j=1}^{s} n_j[b_j] + c[\infty],$$

where $a_i, b_j \in \mathbb{C}$ and $m_i, n_j > 0$. Define

$$h(z) = \frac{\prod_{i=1}^{r}(z - a_i)^{m_i}}{\prod_{j=1}^{s}(z - b_j)^{n_j}}.$$

Then

$$(h) = \sum_{i=1}^{r} m_i[a_i] - \sum_{j=1}^{s} n_j[b_j] - \left(\sum_{i=1}^{r} m_i - \sum_{j=1}^{s} n_j\right)[\infty].$$

Therefore

$$D - (h) = \left(c + \sum_{i=1}^{r} m_i - \sum_{j=1}^{s} n_j\right)[\infty] = (\deg D)[\infty].$$

So $D \sim (\deg D)[\infty]$. □

This theorem is the concrete genus-zero form of the statement that every line bundle on $\mathbb{CP}^1$ is some $\mathcal{O}(n)$.

6 The spaces $L(D)$

For a divisor D on $\mathbb{CP}^1$, define

$$L(D) = \{f \in \mathcal{M}(\mathbb{CP}^1) : (f) + D \geq 0\} \cup \{0\}.$$

Intuitively, $L(D)$ is the space of meromorphic functions whose poles are bounded by D.

Since every divisor is equivalent to $n[\infty]$, it suffices to compute $L(n[\infty])$.

Proposition 6.1. *If $n < 0$, then*

$$L(n[\infty]) = 0.$$

If $n \geq 0$, then

$$L(n[\infty]) = \{\textit{polynomials in } z \textit{ of degree } \leq n\}.$$

In particular,

$$\ell(n[\infty]) = \begin{cases} 0, & n < 0, \\ n+1, & n \geq 0. \end{cases}$$

Proof. Assume first $n \geq 0$. If $f(z) = a_0 + a_1 z + \cdots + a_n z^n$ is a polynomial of degree at most n, then near ∞ we have

$$f(1/w) = a_0 + a_1 w^{-1} + \cdots + a_n w^{-n},$$

so f has a pole of order at most n at ∞ and no other poles. Hence $f \in L(n[\infty])$.

Conversely, let $f \in L(n[\infty])$. Then f has no poles in $\mathbb{C}$ and a pole of order at most n at ∞. Thus f is holomorphic on $\mathbb{C}$, hence entire, and has polynomial growth of order at most n at infinity. Therefore f is a polynomial of degree at most n.

If $n < 0$, then $f \in L(n[\infty])$ would have to vanish at infinity to order at least $-n$, while having no poles anywhere. So f would be a holomorphic function on the compact surface $\mathbb{CP}^1$ vanishing at a point. Since global holomorphic functions on $\mathbb{CP}^1$ are constant, we get $f = 0$. □

Corollary 6.2. For any divisor D on $\mathbb{CP}^1$,

$$\ell(D) = \begin{cases} 0, & \deg D < 0, \\ \deg D + 1, & \deg D \geq 0. \end{cases}$$

Proof. Since $D \sim (\deg D)[\infty]$, the spaces $L(D)$ and $L((\deg D)[\infty])$ are naturally isomorphic by multiplication with a meromorphic function that realizes the equivalence. Hence dimensions agree, and the previous proposition applies. □

7 The line bundles $\mathcal{O}(n)$ on $\mathbb{CP}^1$

The standard line bundle $\mathcal{O}(n)$ can be described by transition function z^n on the overlap $U_0 \cap U_\infty$. A global section is a pair of holomorphic functions (s_0, s_∞) on the two charts satisfying

$$s_\infty(w) = w^n s_0(1/w).$$

If $n \geq 0$, then s_0 must be a polynomial of degree at most n. Hence

$$H^0(\mathbb{CP}^1, \mathcal{O}(n)) \cong \operatorname{span}\{1, z, z^2, \ldots, z^n\}$$

as a vector space on the affine chart. Equivalently, in homogeneous coordinates,

$$H^0(\mathbb{CP}^1, \mathcal{O}(n)) \cong \{\text{homogeneous polynomials of degree } n \text{ in } Z_0, Z_1\}.$$

Thus

$$h^0(\mathbb{CP}^1, \mathcal{O}(n)) = \begin{cases} 0, & n < 0, \\ n+1, & n \geq 0. \end{cases}$$

Remark 7.1. The divisor bundle of $n[\infty]$ is $\mathcal{O}(n)$. So the explicit computation of $L(n[\infty])$ is exactly the explicit computation of global sections of $\mathcal{O}(n)$.

8 The canonical divisor on $\mathbb{CP}^1$

To apply Riemann–Roch we need the canonical divisor K. On the chart U_0 the form dz is holomorphic. Near infinity, write $z = 1/w$. Then

$$dz = d\left(\frac{1}{w}\right) = -\frac{1}{w^2}\, dw.$$

Thus dz has a pole of order 2 at $w = 0$, i.e. at ∞, and no zeros elsewhere. Therefore

$$(dz) = -2[\infty].$$

So a canonical divisor is

$$K \sim -2[\infty],$$

and hence

$$K_{\mathbb{CP}^1} \cong \mathcal{O}(-2).$$

Corollary 8.1. We have

$$\ell(K) = \ell(-2[\infty]) = 0.$$

In particular, $h^0(\mathbb{CP}^1, K) = 0$, as expected for genus 0.

9 Riemann–Roch on $\mathbb{CP}^1$

Now we can verify Riemann–Roch directly.

Theorem 9.1 (Riemann–Roch on $\mathbb{CP}^1$)**.** *For every divisor D on $\mathbb{CP}^1$,*

$$\ell(D) - \ell(K - D) = \deg D + 1.$$

Since $g(\mathbb{CP}^1) = 0$, this is exactly the general Riemann–Roch formula

$$\ell(D) - \ell(K - D) = \deg D + 1 - g.$$

Proof. Write $D \sim n[\infty]$, where $n = \deg D$. Then

$$K - D \sim -2[\infty] - n[\infty] = -(n+2)[\infty].$$

Hence

$$\ell(D) = \ell(n[\infty]), \qquad \ell(K-D) = \ell(-(n+2)[\infty]).$$

We now split into cases.

Case 1: $n \geq 0$**.** Then $\ell(D) = n+1$. Also $-(n+2) < 0$, so $\ell(K-D) = 0$. Thus

$$\ell(D) - \ell(K-D) = (n+1) - 0 = n+1 = \deg D + 1.$$

Case 2: $n = -1$**.** Then $\ell(D) = 0$ and $K - D \sim -[\infty]$, so $\ell(K-D) = 0$. Hence

$$\ell(D) - \ell(K-D) = 0 = \deg D + 1.$$

Case 3: $n \leq -2$**.** Then $\ell(D) = 0$, while

$$K - D \sim -(n+2)[\infty],$$

and now $-(n+2) \geq 0$. Therefore

$$\ell(K-D) = -(n+2) + 1 = -n - 1.$$

So

$$\ell(D) - \ell(K-D) = 0 - (-n-1) = n+1 = \deg D + 1.$$

This proves the theorem. □

10 A useful reformulation with $\mathcal{O}(n)$

Since $K \cong \mathcal{O}(-2)$, Riemann–Roch may be written as

$$h^0(\mathbb{CP}^1, \mathcal{O}(n)) - h^0(\mathbb{CP}^1, \mathcal{O}(-n-2)) = n+1.$$

Because

$$h^0(\mathbb{CP}^1, \mathcal{O}(m)) = \begin{cases} 0, & m < 0, \\ m+1, & m \geq 0, \end{cases}$$

this identity is completely explicit.

By Serre duality one also gets

$$h^1(\mathbb{CP}^1, \mathcal{O}(n)) = h^0(\mathbb{CP}^1, \mathcal{O}(-n-2)).$$

Hence

$$h^1(\mathbb{CP}^1, \mathcal{O}(n)) = \begin{cases} 0, & n \geq -1, \\ -n-1, & n \leq -2. \end{cases}$$

11 What this teaches us

The theorem on $\mathbb{CP}^1$ is elementary enough to compute completely, but it already reveals the general philosophy of Riemann–Roch.

1. **Divisors control allowed poles.** The space $L(D)$ measures meromorphic functions with pole bounds encoded by D.
2. **Degree is the first approximation to dimension.** On $\mathbb{CP}^1$, once $\deg D \geq 0$, the answer is exactly $\deg D + 1$.
3. **The canonical divisor measures the correction term.** On $\mathbb{CP}^1$ the correction is especially simple because $K \sim -2[\infty]$.
4. **Genus zero is rigid.** Every degree-zero divisor is principal, every line bundle is $\mathcal{O}(n)$, and every global meromorphic function is rational.

The general theorem for a compact Riemann surface of genus g keeps exactly the same shape, but the term $\ell(K - D)$ becomes a genuine global obstruction.

12 Thirty exercises

The exercises are arranged in six groups. The first groups are computational, and the later ones synthesize the whole story.

12.1 Group A: coordinates and local orders

Exercise 12.1. Show that the transition map between the two standard affine charts on $\mathbb{CP}^1$ is

$$w = \frac{1}{z}.$$

Explain why ∞ corresponds to $w = 0$.

Exercise 12.2. Compute

$$\operatorname{ord}_\infty(z), \qquad \operatorname{ord}_\infty(1/z), \qquad \operatorname{ord}_\infty(z^m)$$

for $m \in \mathbb{Z}$.

Exercise 12.3. Compute the divisor of $z - a$ for $a \in \mathbb{C}$.

Exercise 12.4. Compute the divisor of

$$\frac{z-a}{z-b}$$

for distinct $a, b \in \mathbb{C}$.

Exercise 12.5. Let

$$P(z) = c \prod_{i=1}^{r} (z - a_i)^{m_i}$$

with $c \neq 0$ and distinct a_i. Compute (P) on $\mathbb{CP}^1$.

12.2 Group B: divisors and linear equivalence on $\mathbb{CP}^1$

Exercise 12.6. Let

$$f(z) = \frac{P(z)}{Q(z)}$$

with coprime polynomials P, Q. Compute (f) in terms of the zeros of P, the zeros of Q, and the order at ∞.

Exercise 12.7. Prove directly that every principal divisor on $\mathbb{CP}^1$ has degree 0.

Exercise 12.8. Let

$$D = \sum_{i=1}^{r} m_i[a_i] - \sum_{j=1}^{s} n_j[b_j]$$

with all $a_i, b_j \in \mathbb{C}$ and $\deg D = 0$. Construct explicitly a rational function h with $(h) = D$.

Exercise 12.9. Let D be any divisor on $\mathbb{CP}^1$. Prove that

$$D \sim (\deg D)[\infty].$$

Exercise 12.10. Deduce that every degree-zero divisor on $\mathbb{CP}^1$ is principal.

12.3 Group C: the spaces $L(D)$

Exercise 12.11. Prove that if $n < 0$, then

$$L(n[\infty]) = 0.$$

Exercise 12.12. Prove that if $n \geq 0$, then

$$L(n[\infty]) = \operatorname{span}_{\mathbb{C}}\{1, z, z^2, \ldots, z^n\}.$$

Exercise 12.13. Compute $\ell(n[\infty])$ for all integers n.

Exercise 12.14. Let $D = [a_1] + \cdots + [a_n]$ with $a_i \in \mathbb{C}$ not necessarily distinct. Show that

$$\ell(D) = n + 1.$$

Hint: first show $D \sim n[\infty]$.

Exercise 12.15. Let D be a divisor on $\mathbb{CP}^1$ of degree d. Show that

$$\ell(D) = 0 \quad \text{if } d < 0, \qquad \ell(D) = d + 1 \quad \text{if } d \geq 0.$$

12.4 Group D: the bundles $\mathcal{O}(n)$

Exercise 12.16. Using the transition function z^n on $U_0 \cap U_\infty$, show that a global section of $\mathcal{O}(n)$ is given by a pair (s_0, s_∞) satisfying

$$s_\infty(w) = w^n s_0(1/w).$$

Exercise 12.17. Show that for $n \geq 0$,

$$H^0(\mathbb{CP}^1, \mathcal{O}(n)) \cong \operatorname{span}_{\mathbb{C}}\{1, z, \ldots, z^n\}$$

on the affine chart U_0.

Exercise 12.18. Show that for $n \geq 0$,

$$H^0(\mathbb{CP}^1, \mathcal{O}(n))$$

may also be identified with homogeneous polynomials of degree n in Z_0, Z_1.

Exercise 12.19. Compute

$$h^0(\mathbb{CP}^1, \mathcal{O}(n))$$

for all integers n.

Exercise 12.20. Show that if $n < 0$, then $\mathcal{O}(n)$ has no nonzero global holomorphic sections.

12.5 Group E: canonical divisor and Riemann–Roch

Exercise 12.21. Compute the divisor of dz on $\mathbb{CP}^1$ and show that

$$K \sim -2[\infty].$$

Exercise 12.22. Compute the divisor of $z^m dz$ for an integer m.

Exercise 12.23. Show that

$$\ell(K) = 0.$$

Interpret this as $h^0(\mathbb{CP}^1, K) = 0$.

Exercise 12.24. Verify Riemann–Roch directly for the divisor $D = n[\infty]$ for every integer n.

Exercise 12.25. Deduce from the previous exercise that Riemann–Roch holds for every divisor D on $\mathbb{CP}^1$.

12.6 Group F: synthesis and applications

Exercise 12.26. Use Riemann–Roch and the formula $K \sim -2[\infty]$ to compute

$$h^1(\mathbb{CP}^1, \mathcal{O}(n))$$

for every integer n.

Exercise 12.27. Show that the degree map induces an isomorphism

$$\mathrm{Pic}(\mathbb{CP}^1) \cong \mathbb{Z}, \qquad \mathcal{O}(n) \longmapsto n.$$

Exercise 12.28. Let D be an effective divisor of degree n on $\mathbb{CP}^1$. Show that there exists a polynomial $P(z)$ of degree exactly n such that the zero divisor of P on $\mathbb{C}$ is exactly $D - n[\infty]$.

Exercise 12.29. Show that the complete linear system $|n[\infty]|$ defines a holomorphic map

$$\mathbb{CP}^1 \longrightarrow \mathbb{P}^n, \qquad [z_0 : z_1] \longmapsto [z_0^n : z_0^{n-1} z_1 : \cdots : z_1^n].$$

Describe its image.

Exercise 12.30. Explain in your own words why $\mathbb{CP}^1$ is the perfect first test case for Riemann–Roch. Your explanation should mention at least the following points:

(a) every meromorphic function is rational,

(b) every divisor class is determined by degree,

(c) $K \sim -2[\infty]$,

(d) the dimension formula becomes completely explicit.

13 Hints and brief solutions

1–5

1. By definition $z = Z_1/Z_0$ on U_0 and $w = Z_0/Z_1$ on U_∞, hence $w = 1/z$. The point $\infty = [0:1]$ satisfies $w = 0$.

2. Since $z = 1/w$ near ∞, we get $\operatorname{ord}_\infty(z) = -1$, $\operatorname{ord}_\infty(1/z) = 1$, and $\operatorname{ord}_\infty(z^m) = -m$.

3. One zero at $z = a$ and one pole at ∞, so $(z - a) = [a] - [\infty]$.

4. Subtract the previous divisors: $((z - a)/(z - b)) = [a] - [b]$.

5. The finite zeros contribute $\sum m_i$, and at infinity there is a pole of order $\sum m_i$. Thus

$$(P) = \sum_{i=1}^{r} m_i[a_i] - \left(\sum_{i=1}^{r} m_i\right)[\infty].$$

6–10

6. Write the finite zeros and poles explicitly. At infinity,

$$\operatorname{ord}_\infty(P/Q) = \deg Q - \deg P.$$

7. Sum the finite orders and the order at infinity; they cancel.

8. Take

$$h(z) = \frac{\prod (z - a_i)^{m_i}}{\prod (z - b_j)^{n_j}}.$$

Degree zero guarantees no extra order at infinity.

9. Construct h from all finite points as in the proof in the note. Then $D - (h)$ is supported only at ∞, and its coefficient is $\deg D$.

10. If $\deg D = 0$, then the previous exercise gives $D \sim 0$, hence $D = (h)$.

11–15

11. A function in $L(n[\infty])$ would have no poles anywhere and would have to vanish at infinity. Global holomorphic functions on $\mathbb{CP}^1$ are constant, so only 0 works.

12. A function in $L(n[\infty])$ is entire on $\mathbb{C}$ and has pole order at most n at infinity, hence is a polynomial of degree at most n.

13. The answer is 0 for $n < 0$ and $n + 1$ for $n \geq 0$.

14. Since each $[a_i] \sim [\infty]$, we get $D \sim n[\infty]$, so $\ell(D) = \ell(n[\infty]) = n + 1$.

15. Use $D \sim d[\infty]$ where $d = \deg D$.

16–20

16. This is exactly the cocycle condition for sections of a line bundle with transition function z^n.

17. Holomorphicity of $s_\infty(w) = w^n s_0(1/w)$ at $w = 0$ forces s_0 to be a polynomial of degree at most n.

18. A homogeneous degree-n polynomial

$$F(Z_0, Z_1) = \sum_{k=0}^{n} a_k Z_0^{n-k} Z_1^k$$

becomes the polynomial $\sum a_k z^k$ on U_0.

19. Thus $h^0(\mathbb{CP}^1, \mathcal{O}(n)) = 0$ for $n < 0$ and $n + 1$ for $n \geq 0$.

20. If a nonzero global section existed for $n < 0$, it would correspond to a divisor of negative degree but effective divisor of zeros, impossible.

21–25

21. Use $z = 1/w$ and compute

$$dz = -w^{-2}\, dw.$$

Hence $(dz) = -2[\infty]$.

22. Combine $(z^m) = m[0] - m[\infty]$ with $(dz) = -2[\infty]$ to get

$$(z^m dz) = m[0] - (m+2)[\infty].$$

23. Since $K \sim -2[\infty]$ has negative degree, $\ell(K) = 0$.

24. Do the three cases $n \geq 0$, $n = -1$, and $n \leq -2$.

25. Since every divisor is equivalent to $n[\infty]$, the previous exercise is enough.

26–30

26. Use

$$h^0(\mathcal{O}(n)) - h^1(\mathcal{O}(n)) = n + 1$$

together with $h^1(\mathcal{O}(n)) = h^0(\mathcal{O}(-n-2))$ or solve case by case.

27. Every line bundle is $\mathcal{O}(D)$ for some divisor D, and $D \sim (\deg D)[\infty]$. Thus $\mathcal{O}(D) \cong \mathcal{O}(\deg D)$.

28. Take the monic polynomial

$$P(z) = \prod (z - a_i)^{m_i}$$

where $D = \sum m_i[a_i]$ on the affine chart; its divisor is $D - n[\infty]$.

29. The image is the rational normal curve of degree n in $\mathbb{P}^n$.

30. A good answer should stress that on $\mathbb{CP}^1$ every ingredient of Riemann–Roch can be computed by hand.

14 Final summary

For $\mathbb{CP}^1$ the Riemann–Roch theorem is not mysterious at all. Everything boils down to four explicit facts:

1. $\mathcal{M}(\mathbb{CP}^1) = \mathbb{C}(z)$,
2. every divisor satisfies $D \sim (\deg D)[\infty]$,
3. $L(n[\infty])$ is the space of polynomials of degree at most n,
4. $K \sim -2[\infty]$.

Therefore

$$\ell(D) - \ell(K - D) = \deg D + 1$$

becomes a direct polynomial count.

This is exactly why the genus-zero case should be mastered first: it turns the general theorem into a sequence of concrete computations with rational functions, divisors, and the point at infinity.